Supplementary Table S1

Inputs to the production systems ha^{-1}in eachyear during the study period

Production systems	2009–10	2010–11	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19	Total
Gmelina											
N (kg)	4.2	4.2	4.2	4.2	-*	-	-	-	-	-	16.8
P_2O_5 (kg)	4.2	4.2	4.2	4.2	-	-	-	-	-	-	16.8
K_2O (kg)	4.2	4.2	4.2	4.2	-	-	-	-	-	-	16.8
FYM (t)	2.0	2.0	2.0	2.0	-	-	-	-	-	-	8.0
Pesticides (kg/l)	5.0	5.0	5.0	5.0	-	-	-	-	-	-	20.0
Diesel (l)	55.4	51.6	51.6	51.6	51.6	51.6	51.6	51.6	51.6	51.6	520.0
Manpower (h)	136.0	87.5	80.0	24.0	15.5	15.5	11.5	11.5	11.5	67.0	460.0
Mango											
N (kg)	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0	650.0
P_2O_5 (kg)	5.0	10.0	20.0	30.0	45.0	60.0	60.0	60.0	60.0	60.0	410.0
K_2O (kg)	10.0	20.0	30.0	40.0	50.0	80.0	90.0	100.0	110.0	120.0	650.0
FYM (t)	0.5	1.0	1.5	2.0	2.5	4.0	4.5	5.5	5.5	5.5	32.5
Zinc (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.4
Boron (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.1
Pesticides (kg l-1)	1.3	1.3	1.3	6.9	6.9	6.9	6.9	6.9	6.9	6.9	52.1
Diesel (l)	11.3	56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5	519.8
Manpower (h)	72.0	54.0	54.0	49.5	49.5	49.5	49.5	49.5	49.5	100.0	577.0
Pigeon pea											
N (kg)	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0	220.0
P_2O_5 (kg)	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	190.0
K_2O (kg)	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	19.0	190.0
Pesticides (kg l-1)	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	8.0
Diesel (l)	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	670.0
Manpower (h)	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	50.0	500.0
Gmelina-Mango											
N (kg)	21.1	31.1	41.1	51.1	60.0	70.0	80.0	90.0	100.0	110.0	654.2
P_2O_5 (kg)	6.1	11.1	21.1	31.1	45.0	60.0	60.0	60.0	60.0	60.0	414.2
K_2O (kg)	11.1	21.1	31.1	41.1	50.0	80.0	90.0	100.0	110.0	120.0	654.2
FYM (t)	1.0	1.5	2.0	2.5	2.5	4.0	4.5	5.5	5.5	5.5	34.5
Zinc (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.2
Boron (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.1
Pesticides (kg/l)	5.0	5.0	5.0	10.7	6.9	6.9	6.9	6.9	6.9	6.9	67.1
Diesel (l)	11.3	56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5	56.5	519.8
Manpower (h)	106.0	75.9	134.0	73.5	65.0	65.0	61.0	61.0	61.0	167.0	869.4
Gmelina-Pigeon pea											
N (kg)	26.2	26.2	26.2	26.2	22.0	22.0	22.0	22.0	22.0	22.0	236.8
P_2O_5 (kg)	23.2	23.2	23.2	23.2	19.0	19.0	19.0	19.0	19.0	19.0	206.8
K_2O (kg)	23.2	23.2	23.2	23.2	19.0	19.0	19.0	19.0	19.0	19.0	206.8

FYM (t)	2.0	2.0	2.0	2.0	-	-	-	-	-	-	8.0
Pesticides (kg l-1)	5.8	5.8	5.8	5.8	0.8	0.8	0.8	0.8	0.8	0.8	28.0
Diesel (l)	55.4	118.6	118.6	118.6	118.6	118.6	118.6	118.6	118.6	118.6	1123.0
Manpower (h)	186.0	137.5	130.0	74.0	65.5	65.5	61.5	61.5	61.5	117.0	960.0
Mango-Pigeon pea											
N (kg)	42.0	52.0	62.0	72.0	82.0	92.0	102.0	112.0	122.0	132.0	870.0
P_2O_5 (kg)	24.0	29.0	39.0	49.0	64.0	79.0	79.0	79.0	79.0	79.0	600.0
K_2O (kg)	29.0	39.0	49.0	59.0	69.0	99.0	109.0	119.0	129.0	139.0	840.0
FYM (t)	0.5	1.0	1.5	2.0	2.5	4.0	4.5	5.5	5.5	5.5	32.5
Zinc (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.4
Boron (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.1
Pesticides (kg l-1)	2.1	2.1	2.1	7.7	7.7	7.7	7.7	7.7	7.7	7.7	60.1
Diesel (l)	11.3	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5	1122.8
Manpower (h)	120.0	100.0	100.0	99.5	99.5	99.5	99.5	99.5	99.5	150.0	1067.0
Gmelina-Mango-Pigeon pea											
N (kg)	43.1	53.1	63.1	73.1	82.0	92.0	102.0	112.0	122.0	132.0	874.2
P_2O_5 (kg)	25.1	30.1	40.1	50.1	64.0	79.0	79.0	79.0	79.0	79.0	604.2
K_2O (kg)	30.1	40.1	50.1	60.1	69.0	99.0	109.0	119.0	129.0	139.0	844.2
FYM (t)	1.0	1.5	2.0	2.5	2.5	4.0	4.5	5.5	5.5	5.5	34.5
Zinc (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.4
Boron (kg)	-	-	-	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.1
Pesticides (kg l-1)	6.0	6.0	6.0	12.0	8.0	8.0	8.0	8.0	8.0	8.0	78.0
Diesel (l)	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	67.0	670.0
Manpower (h)	125.0	104.0	104.0	104.0	104.0	104.0	104.0	104.0	104.0	155.0	1112.0

*Blank cell with “-“sign indicates no inputs for the year

Supplementary Table S2

Prevalent market price, energy equivalents, and carbon emission factor of the inputs and outputs of production systems underthe study

Input/Output	Unit	Prevalent market price (USD unit^{-1}) during 2009-10 to 2018-19										Energy equivalent (MJ unit^{-1})	C emissionfactor
		2009-10	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19		
Inputs													
Pigeon pea seeds	kg	1.00	1.08	1.12	1.22	1.41	1.30	1.36	1.56	1.47	1.60	14.7	-
Mango seedlings (dry weight basis)	kg	0.37	0.40	0.41	0.45	0.52	0.48	0.50	0.57	0.54	0.59	12.5	-
Gmelina seedlings (dry weight basis)	kg	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	12.5	-
Fertiliser N	kg	0.15	0.15	0.14	0.15	0.17	0.18	0.28	0.45	0.51	0.71	60.60	6.92
Fertiliser P_2O_5	kg	0.10	0.10	0.09	0.09	0.11	0.11	0.18	0.29	0.33	0.46	11.1	1.66
Fertiliser K_2O	kg	0.10	0.09	0.09	0.09	0.11	0.11	0.18	0.29	0.33	0.46	6.70	1.47
FYM	kg	0.03	0.03	0.03	0.03	0.04	0.04	0.04	0.04	0.05	0.06	0.03	-
Zn	kg	28.1	26.2	26.2	24.6	26.3	24.9	25.0	27.3	24.3	26.8	20.9	0.43
B	kg	15.8	13.5	13.5	12.5	13.1	12.1	12.2	13.1	11.8	12.7	5.29	2.03
Diesel	L	1.12	1.22	1.36	1.12	1.30	1.05	0.97	1.05	1.04	1.04	56.31	2.68
Manpower	8 h d^{-1}	3.77	3.33	3.56	3.61	3.83	3.54	3.56	3.95	3.46	3.82	15.68	-
Triazophos	L	6.06	5.35	5.73	5.80	6.15	5.70	5.72	6.35	5.56	6.14	120	8.87
Chlorpyriphos	L	7.27	6.42	6.87	6.96	7.38	6.84	6.86	7.62	6.67	7.37	7.56	0.56
Acephate + Imidacloprid	kg	11.5	10.2	10.9	11.0	11.7	10.8	10.9	12.1	10.6	11.7	33.27	2.46
Sulphur	kg	1.38	1.23	1.32	1.33	1.41	1.31	1.32	1.46	1.28	1.41	120	8.87
Imidacloprid	L	11.4	10.2	10.9	11.0	11.7	10.8	10.9	12.1	10.6	11.7	120	0.78

Carbendazim	kg	10.0	8.9	9.5	9.6	10.2	9.5	9.5	10.5	9.2	10.2	10.59	8.87
Mancozeb	kg	10.0	8.9	9.5	9.6	10.2	9.5	9.5	10.5	9.2	10.2	120	0.78
Outputs													
Gmelina foliage and twigs (pruned)	kg	0.09	0.08	0.08	0.08	0.09	0.08	0.08	0.09	0.08	0.09	19.07	8.87
Gmelina bole	m^{-3}	263	233	249	252	267	247	248	276	241	267	18.26	
Gmelina branches	kg	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	17.13	
Mango foliage and twigs (pruned)	kg	0.09	0.08	0.08	0.08	0.09	0.08	0.08	0.09	0.08	0.09	19.18	
Mango fruits	kg	0.70	0.62	0.66	0.67	0.71	0.66	0.66	0.74	0.64	0.71	23.0	
Mango fuelwood	kg	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	16.89	
Harvested pigeon pea seeds	kg	0.53	0.47	0.50	0.50	0.53	0.49	0.50	0.55	0.48	0.53	14.7	
Harvested pigeon pea strove	kg	0.09	0.08	0.08	0.08	0.09	0.08	0.08	0.09	0.08	0.09	12.5	

Supplementary Table S3

Economic analysis of production systems under study between 2009-10 and 2018-19

Production systems	Attributes	Year									
		2009–10	2010–11	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19
Gmelina	COC (USD ha^{-1})	544	390	369	208	107	98	83	81	78	74
	GR (USD ha^{-1})	3	5	11	21	38	53	63	73	78	5955
	BCR	-0.99	-0.99	-0.97	-0.90	-0.64	-0.46	-0.24	-0.11	0.00	79.76
Mango	COC (USD ha^{-1})	288	244	260	294	296	295	305	325	346	376
	GR (USD ha^{-1})	0	1	1	661	1142	1318	1998	1537	1502	1264
	BCR	-1.00	-1.00	-1.00	1.25	2.86	3.47	5.55	3.73	3.35	2.36
Pigeon pea	COC (USD ha^{-1})	222	221	221	222	211	194	183	180	168	164
	GR (USD ha^{-1})	662	704	652	753	625	624	488	469	395	383
	BCR	1.99	2.19	1.95	2.39	1.96	2.23	1.67	1.61	1.35	1.34
Gmelina- mango	COC (USD ha^{-1})	488	468	462	453	395	386	391	409	424	451
	GR (USD ha^{-1})	1	5	11	524	1072	1211	1986	1438	1436	4928
	BCR	-1.00	-0.99	-0.98	0.16	1.72	2.14	4.08	2.52	2.38	9.92
Gmelina- pigeon pea	COC (USD ha^{-1})	757	606	584	426	313	288	262	258	243	235
	GR (USD ha^{-1})	679	660	601	644	388	360	262	210	153	6346
	BCR	-0.10	0.09	0.03	0.51	0.24	0.25	0.00	-0.18	-0.37	26.00
Mango - pigeon pea	COC (USD ha^{-1})	501	459	476	512	502	485	484	502	511	537
	GR (USD ha^{-1})	637	637	577	1288	1506	1720	2300	1864	1767	1504
	BCR	0.27	0.39	0.21	1.52	2.00	2.55	3.75	2.72	2.46	1.80
Gmelina- mango - pigeon pea	COC (USD ha^{-1})	632	639	633	604	538	517	516	533	542	567
	GR (USD ha^{-1})	603	590	593	1018	1379	1538	2173	1727	1587	5171
	BCR	-0.05	-0.08	-0.06	0.69	1.56	1.97	3.21	2.24	1.93	8.12

Discount rate = 12%

Supplementary Table S4

Sensitivity of economic parameters to different discount rates of different production systems

Economic parameters	Production systems						
	Gmelina	Mango	Pigeon pea	*Gmelina* – mango	*Gmelina* - pigeon pea	Mango - pigeon pea	*Gmelina* - mango - pigeon pea
Discount rate = 8%							
Net present value (USD ha^{-1})	4889^{d}	6731^{c}	3748^{d}	8967^{b}	6882^{c}	9119^{b}	11283^{a}
Benefit-cost ratio	2.83^{a}	2.51^{b}	2.17^{d}	2.37^{c}	2.03^{e}	2.10^{e}	2.26^{d}
Land expectation value (USD ha^{-1})	9107^{d}	12538^{c}	6983^{e}	16705^{b}	12821^{c}	16988^{b}	21020^{a}
Annual equivalent value (USD ha^{-1})	729^{d}	1003^{c}	559^{d}	1336^{b}	1026^{c}	1359^{b}	1682^{a}
Discount rate = 10%							
e	3880^{e}	5813^{d}	3427^{e}	7532^{c}	5754^{d}	8029^{b}	9686^{a}
Benefit-cost ratio	2.41^{a}	2.42^{a}	2.19^{b}	2.21^{b}	1.85^{d}	2.06^{c}	2.15^{b}
Land expectation value (USD ha^{-1})	6314^{e}	9460^{d}	5578^{d}	12257^{c}	9365^{d}	13067^{b}	15764^{a}
Annual equivalent value (USD ha^{-1})	631^{e}	946^{d}	558^{f}	1226^{c}	936^{d}	1307^{b}	1576^{a}
Discount rate =14%							
Net present value (USD ha^{-1})	2395^{e}	4366^{d}	2899^{e}	5335^{c}	4067^{d}	6293^{b}	7219^{a}
Benefit-cost-ratio	1.71^{c}	2.23^{a}	2.22^{a}	1.90^{b}	1.54^{d}	1.96^{b}	1.94^{b}
Land expectation value (USD ha^{-1})	3280^{f}	5979^{d}	3969^{e}	7306^{c}	5569^{d}	8618^{b}	9886^{a}
Annual equivalent value (USD ha^{-1})	459^{g}	837^{d}	556^{f}	1023^{c}	780^{e}	1207^{b}	1384^{a}
Discount rate = 16%							
Net present value (USD ha^{-1})	1852^{f}	3795^{d}	2680^{e}	4496^{c}	3437^{d}	5601^{b}	6266^{a}
Benefit-cost-ratio	1.41^{c}	2.14^{a}	2.23^{a}	1.75^{b}	1.40^{c}	1.91^{ab}	1.84^{b}
Land expectation value (USD ha^{-1})	2395^{f}	4908^{d}	3466^{e}	5814^{c}	4444^{d}	7243^{b}	8102^{a}
Annual equivalent value (USD ha^{-1})	383^{g}	785^{d}	555^{f}	930^{c}	711^{e}	1159^{b}	1296^{a}

Means followed by similar letters (Tukey's ranking) within a row do not differ at 5% level of significance

Supplementary Table S5

Mean dry biomass (t ha^{-1}) of *Gmelina* litter, mango litter and pigeon pea residue recycled under different production systems during 2009-10 to 2018-19

Production systems	Attributes	Year										
		2009–10	2010–11	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19	Mean
Gmelina	GLR	0.13	0.26	0.39	0.76	1.23	3.01	5.33	7.89	11.25	13.45	4.37
Mango	MLR	0.01	0.02	0.05	0.13	0.34	0.79	0.91	1.97	1.99	2.62	0.88
Pigeon pea	PR	3.65	3.82	3.60	3.91	3.42	3.59	3.36	2.03	2.94	2.66	3.30
Gmelina- mango	GLR	0.10	0.20	0.27	0.50	0.80	1.99	3.35	5.01	7.48	8.30	2.80
	MLR	0.01	0.02	0.05	0.13	0.34	0.79	0.91	1.97	1.99	2.62	0.88
Gmelina- pigeon pea	GLR	0.13	0.25	0.40	0.86	1.33	3.04	5.88	8.92	12.17	15.02	4.80
	PR	3.01	3.36	3.01	3.19	2.65	3.20	1.99	1.49	1.13	0.96	2.40
Mango - pigeon pea	MLR	0.01	0.02	0.08	0.20	0.55	0.98	1.16	2.57	2.52	2.55	1.06
	PR	3.00	2.90	2.68	3.03	2.28	2.54	2.89	2.43	1.76	1.70	2.52
Gmelina- mango - pigeon pea	GLR	0.09	0.17	0.27	0.58	0.90	1.93	3.92	6.53	8.26	9.85	3.25
	MLR	0.01	0.02	0.08	0.19	0.52	0.93	1.11	2.00	2.12	2.02	0.90
	PR	3.01	3.11	2.76	2.73	2.12	2.36	1.49	1.88	1.12	0.95	2.15

GLR, MLR, and PR are the *Gmelina*litter residue, mango litter residue, and pigeon pea residue, respectively.

Supplementary Table S6

Energy attributes (MJ ha^{-1}) of different production systems during 2009-10 to 2018-19

Year	Production system	Man Power	Seed/ seedling	FYM	N	P	K	Zn	B	Diesel	Pesticide	Non-renewable Input	Total input	Output
2009–10	*Gmelin*a	2132	1250	60	255	47	28	0	0	845	39	3442	4656	908
	Mango	1129	313	15	606	56	67	0	0	845	29	1456	3058	100
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	61815
	Gmelina- mango	1693	1250	60	797	90	88	0	0	845	65	3003	4889	443
	Gmelina- pigeon pea	3073	1544	60	1467	235	142	0	0	1126	129	4677	7777	59193
	Mango - pigeon pea	2070	607	15	1818	244	181	0	0	1126	119	2691	6179	50387
	Gmelina- mango - pigeon pea	2281	1544	60	2009	279	202	0	0	1126	155	3885	7657	53573
2010–11	*Gmelina*	1372	0	60	255	47	28	0	0	422	39	1432	2223	1465
	Mango	847	0	30	1212	111	134	0	0	422	29	877	2785	162
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	66080
	Gmelina- mango	1529	0	75	1403	146	155	0	0	422	65	1604	3795	1398
	Gmelina- pigeon pea	2313	294	60	1467	235	142	0	0	985	129	2667	5625	61392
	Mango - pigeon pea	1788	294	30	2424	300	248	0	0	985	119	2112	6187	54346
	Gmelina- mango - pigeon pea	2234	294	75	2615	335	269	0	0	985	155	2603	6962	55737
2011–12	*Gmelina*	1254	0	60	255	47	28	0	0	422	39	1314	2105	3478
	Mango	847	0	45	1818	222	201	0	0	422	29	892	3584	386
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	61813
	Gmelina- mango	1411	0	90	2009	257	222	0	0	422	65	1501	4477	3316
	Gmelina- pigeon pea	2195	294	60	1467	235	142	0	0	985	129	2549	5508	57931
	Mango - pigeon pea	1788	294	45	3030	411	315	0	0	985	119	2127	6986	49909
	Gmelina- mango - pigeon pea	2117	294	90	3221	446	336	0	0	985	155	2501	7644	54075

2012–13	*Gmelina*	376	0	60	255	47	28	0	0	422	39	436	1227	6217
	Mango	776	0	60	2424	333	268	4	1	422	450	836	4738	30586
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	68381
	Gmelina- mango	1090	0	105	2615	368	289	4	1	422	487	1195	5380	28200
	Gmelina- pigeon pea	1317	294	60	1467	235	142	0	0	985	129	1671	4630	63111
	Mango - pigeon pea	1717	294	60	3636	522	382	4	1	985	540	2071	8141	85352
	Gmelina- mango - pigeon pea	1678	294	105	3827	557	403	4	1	985	577	2077	8430	75324
2013–14	*Gmelina*	243	0	0	0	0	0	0	0	422	0	243	665	12396
	Mango	776	0	75	3030	444	335	4	1	422	447	851	5533	57419
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	60303
	Gmelina- mango	1090	0	75	3030	444	335	4	1	422	447	1165	5847	59600
	Gmelina- pigeon pea	1184	294	0	1212	189	114	0	0	985	90	1478	4068	51788
	Mango - pigeon pea	1717	294	75	4242	633	449	4	1	985	537	2086	8936	95632
	Gmelina- mango - pigeon pea	1678	294	75	4242	633	449	4	1	985	537	2047	8897	94074
2014–15	*Gmelina*	243	0	0	0	0	0	0	0	422	0	243	665	18422
	Mango	776	0	90	3636	555	402	4	1	422	447	866	6332	72054
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	64082
	Gmelina- mango	1090	0	90	3636	555	402	4	1	422	447	1180	6646	76217
	Gmelina- pigeon pea	1184	294	0	1212	189	114	0	0	985	90	1478	4068	63092
	Mango - pigeon pea	1717	294	90	4848	744	516	4	1	985	537	2101	9735	116420
	Gmelina- mango - pigeon pea	1678	294	90	4848	744	516	4	1	985	537	2062	9696	115094
2015–16	*Gmelina*	180	0	0	0	0	0	0	0	422	0	180	603	23662
	Mango	776	0	105	4242	555	469	4	1	422	447	881	7020	116852
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	47272
	Gmelina- mango	1090	0	105	4242	555	469	4	1	422	447	1195	7334	127822
	Gmelina- pigeon pea	1121	294	0	1212	189	114	0	0	985	90	1415	4005	54466
	Mango - pigeon pea	1717	294	105	5454	744	583	4	1	985	537	2116	10423	151190
	Gmelina- mango - pigeon pea	1678	294	105	5454	744	583	4	1	985	537	2077	10384	154632
2016–17	*Gmelina*	180	0	0	0	0	0	0	0	422	0	180	603	28330
	Mango	776	0	120	4848	555	536	4	1	422	447	896	7708	94413

	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	43842
	Gmelina- mango	1090	0	120	4848	555	536	4	1	422	447	1210	8022	102151
	Gmelina- pigeon pea	1121	294	0	1212	189	114	0	0	985	90	1415	4005	50291
	Mango - pigeon pea	1717	294	120	6060	744	650	4	1	985	537	2131	11111	136271
	Gmelina- mango - pigeon pea	1678	294	120	6060	744	650	4	1	985	537	2092	11072	139164
2017–18	*Gmelina*	180	0	0	0	0	0	0	0	422	0	180	603	33647
	Mango	776	0	135	5454	555	603	4	1	422	447	911	8396	102322
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	31276
	Gmelina- mango	1090	0	135	5454	555	603	4	1	422	447	1225	8710	114072
	Gmelina- pigeon pea	1121	294	0	1212	189	114	0	0	985	90	1415	4005	48350
	Mango - pigeon pea	1717	294	135	6666	744	717	4	1	985	537	2146	11799	145859
	Gmelina- mango - pigeon pea	1678	294	135	6666	744	717	4	1	985	537	2107	11760	136893
2018–19	*Gmelina*	180	0	0	0	0	0	0	0	422	0	180	603	952907
	Mango	776	0	150	6060	555	670	4	1	422	447	926	9084	310932
	Pigeon pea	941	294	0	1212	189	114	0	0	845	90	1235	3684	28653
E	*Gmelina*- mango	1090	0	150	6060	555	670	4	1	422	447	1240	9398	929744
	Gmelina- pigeon pea	1121	294	0	1212	189	114	0	0	985	90	1415	4005	1002092
	Mango - pigeon pea	1717	294	150	7272	744	784	4	1	985	537	2161	12487	360326
	Gmelina- mango - pigeon pea	1678	294	150	7272	744	784	4	1	985	537	2122	12448	894262

Table S7

Greenhouse gassesemission (kg CO_2 eq. ha^{-1})fromdifferent production systems (2009-10 to 2018-19)

Year	Production system	Source						Total GHG emission
		N	P	K	Micronutrient	Diesel	Pesticide	
2009–10	*Gmelina*	29.1	7.0	6.2	0.0	40.2	2.9	85.3
	Mango	69.2	8.3	14.7	0.0	40.2	2.1	134.5
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	91.0	13.5	19.3	0.0	40.2	4.8	168.9
	Gmelina-pigeon pea	167.5	35.2	31.2	0.0	53.6	9.6	297.0
	Mango-pigeon pea	207.6	36.5	39.7	0.0	53.6	8.8	346.2
	Gmelina-mango-pigeon pea	229.4	41.7	44.3	0.0	53.6	11.5	380.5
2010–11	*Gmelina*	29.1	7.0	6.2	0.0	20.1	2.9	65.2
	Mango	138.4	16.6	29.4	0.0	20.1	2.1	206.6
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	160.2	21.8	34.0	0.0	20.1	4.8	241.0
	Gmelina-pigeonpea	167.5	35.2	31.2	0.0	46.9	9.6	290.3
	Mango-pigeon pea	276.8	44.8	54.4	0.0	46.9	8.8	431.7
	Gmelina- mango-pigeon pea	298.6	50.0	59.0	0.0	46.9	11.5	466.0
2011–12	*Gmelina*	29.1	7.0	6.2	0.0	20.1	2.9	65.2
	Mango	207.6	33.2	44.1	0.0	20.1	2.1	307.1
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	229.4	38.4	48.7	0.0	20.1	4.8	341.5

	Gmelina-pigeon pea	167.5	35.2	31.2	0.0	46.9	9.6	290.3
	Mango-pigeon pea	346.0	61.4	69.1	0.0	46.9	8.8	532.2
	Gmelina-mango-pigeon pea	367.8	66.6	73.7	0.0	46.9	11.5	566.5
2012–13	*Gmelina*	29.1	7.0	6.2	0.0	20.1	2.9	65.2
	Mango	276.8	49.8	58.8	0.4	20.1	33.3	439.2
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	298.6	55.0	63.4	0.4	20.1	36.0	473.5
	Gmelina-pigeon pea	167.5	35.2	31.2	0.0	46.9	9.6	290.3
	Mango-pigeon pea	415.2	78.0	83.8	0.4	46.9	39.9	664.2
	Gmelina- mango-pigeon pea	437.0	83.2	88.4	0.4	46.9	42.6	698.6
2013–14	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	346.0	66.4	73.5	0.4	20.1	33.0	539.4
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	346.0	66.4	73.5	0.4	20.1	33.0	539.4
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2
	Mango-pigeon pea	484.4	94.6	98.5	0.4	46.9	39.7	764.5
	Gmelina- mango-pigeon pea	484.4	94.6	98.5	0.4	46.9	39.7	764.5
2014–15	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	415.2	83.0	88.2	0.4	20.1	33.0	639.9
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	415.2	83.0	88.2	0.4	20.1	33.0	639.9
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2

	Mango-pigeon pea	553.6	111.2	113.2	0.4	46.9	39.7	865.0
	Gmelina- mango-pigeon pea	553.6	111.2	113.2	0.4	46.9	39.7	865.0
2015–16	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	484.4	83.0	102.9	0.4	20.1	33.0	723.8
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	484.4	83.0	102.9	0.4	20.1	33.0	723.8
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2
	Mango-pigeon pea	622.8	111.2	127.9	0.4	46.9	39.7	948.9
	Gmelina- mango-pigeon pea	622.8	111.2	127.9	0.4	46.9	39.7	948.9
2016–17	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	553.6	83.0	117.6	0.4	20.1	33.0	807.7
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	553.6	83.0	117.6	0.4	20.1	33.0	807.7
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2
	Mango-pigeon pea	692.0	111.2	142.6	0.4	46.9	39.7	1032.8
	Gmelina- mango-pigeon pea	692.0	111.2	142.6	0.4	46.9	39.7	1032.8
2017–18	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	622.8	83.0	132.3	0.4	20.1	33.0	891.6
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	622.8	83.0	132.3	0.4	20.1	33.0	891.6
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2
	Mango-pigeon pea	761.2	111.2	157.3	0.4	46.9	39.7	1116.7

	Gmelina- mango-pigeon pea	761.2	111.2	157.3	0.4	46.9	39.7	1116.7
2018–19	*Gmelina*	0.0	0.0	0.0	0.0	20.1	0.0	20.1
	Mango	692.0	83.0	147.0	0.4	20.1	33.0	975.5
	Pigeon pea	138.4	28.2	25.0	0.0	40.2	6.7	238.5
	Gmelina-mango	692.0	83.0	147.0	0.4	20.1	33.0	975.5
	Gmelina-pigeon pea	138.4	28.2	25.0	0.0	46.9	6.7	245.2
	Mango-pigeon pea	830.4	111.2	172.0	0.4	46.9	39.7	1200.6
	Gmelina-mango-pigeon pea	830.4	111.2	172.0	0.4	46.9	39.7	1200.6

Supplementary Table S8

Soil physicochemical characteristicsunder different production systems during the period of study

Attribute		Year										
		Initial	2009–10	2010–11	2011–12	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19
pH	*Gmelina*	5.1	5.2	5.2	5.3	5.3	5.4	5.5	5.5	5.6	5.7	5.8
	Mango	5.1	5.2	5.2	5.3	5.3	5.4	5.5	5.5	5.6	5.7	5.8
	Pigeon pea	5.3	5.4	5.5	5.5	5.6	5.6	5.7	5.7	5.8	5.8	5.9
	Gmelina- mango	5.2	5.3	5.3	5.4	5.4	5.5	5.5	5.6	5.6	5.7	5.8
	Gmelina- pigeon pea	5.2	5.3	5.3	5.4	5.4	5.5	5.5	5.6	5.6	5.7	5.8
	Mango - pigeon pea	5.2	5.3	5.3	5.4	5.4	5.5	5.5	5.6	5.6	5.7	5.8
	Gmelina- mango - pigeon pea	5.5	5.6	5.6	5.7	5.7	5.8	5.8	5.9	6.1	6.3	6.5
OC (%)	*Gmelina*	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4
	Mango	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4
	Gmelina- mango	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4
	Pigeon pea	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5
	Gmelina- pigeon pea	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.5
	Mango - pigeon pea	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5
	Gmelina- mango - pigeon pea	0.3	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5	0.6	0.6
Available N (kg ha^{-1})	*Gmelina*	144.2	146.1	147.9	149.8	151.6	153.5	155.3	157.2	159.0	160.9	162.7
	Mango	140.2	143.2	146.3	149.3	152.4	155.4	158.4	161.5	164.5	167.6	170.6
	Pigeon pea	144.2	149.1	154.0	159.0	163.9	168.8	173.7	178.6	183.6	188.5	193.4
	Gmelina- mango	148.2	152.2	156.2	160.2	164.2	168.2	172.1	176.1	180.1	184.1	188.1
	Gmelina- pigeon pea	144.2	149.8	155.5	161.1	166.7	172.4	178.0	183.6	189.2	194.9	200.5

	Mango - pigeon pea	144.2	152.4	160.6	168.9	177.1	185.3	193.5	201.7	210.0	218.2	226.4
	Gmelina- mango - pigeon pea	142.7	153.5	164.2	175.0	185.8	196.6	207.3	218.1	228.9	239.6	250.4
Available P (kg ha^{-1})	*Gmelina*	9.7	10.3	10.9	11.4	12.0	12.6	13.2	13.8	14.3	14.9	15.5
	Mango	9.7	10.4	11.1	11.8	12.5	13.3	14.0	14.7	15.4	16.1	16.8
	Pigeon pea	9.7	11.1	12.5	13.9	15.3	16.7	18.0	19.4	20.8	22.2	23.6
	Gmelina- mango	9.8	11.1	12.3	13.6	14.8	16.1	17.4	18.6	19.9	21.1	22.4
	Gmelina- pigeon pea	9.7	11.3	12.9	14.5	16.1	17.7	19.3	20.9	22.5	24.1	25.7
	Mango - pigeon pea	9.7	11.7	13.6	15.6	17.5	19.5	21.4	23.4	25.3	27.3	29.2
	Gmelina- mango - pigeon pea	10.5	12.7	14.9	17.1	19.3	21.5	23.7	25.9	28.1	30.3	32.5
Available K (kg ha^{-1})	*Gmelina*	152.5	154.6	156.7	158.8	160.9	163.0	165.0	167.1	169.2	171.3	173.4
	Mango	151.6	154.6	157.6	160.5	163.5	166.5	169.5	172.5	175.4	178.4	181.4
	Pigeon pea	152.5	156.6	160.6	164.7	168.7	172.8	176.8	180.9	184.9	189.0	193.0
	Gmelina- mango	156.0	158.9	161.8	164.6	167.5	170.4	173.3	176.2	179.0	181.9	184.8
	Gmelina- pigeon pea	152.5	158.7	165.0	171.2	177.4	183.7	189.9	196.1	202.3	208.6	214.8
	Mango - pigeon pea	152.5	159.1	165.7	172.3	178.9	185.6	192.2	198.8	205.4	212.0	218.6
	Gmelina- mango - pigeon pea	153.4	160.6	167.7	174.9	182.0	189.2	196.4	203.5	210.7	217.8	225.0
Available Fe (mg kg^{-1})	*Gmelina*	27.2	28.5	29.7	31.0	32.2	33.5	34.8	36.0	37.3	38.5	39.8
	Mango	27.3	29.1	30.8	32.6	34.3	36.1	37.8	39.6	41.3	43.1	44.8
	Pigeon pea	27.1	30.3	33.5	36.8	40.0	43.2	46.4	49.6	52.9	56.1	59.3
	Gmelina- mango	27.3	29.6	31.9	34.2	36.5	38.8	41.1	43.4	45.7	48.0	50.3
	Gmelina- pigeon pea	26.9	30.6	34.2	37.9	41.6	45.3	48.9	52.6	56.3	59.9	63.6
	Mango - pigeon pea	27.1	32.0	36.9	41.7	46.6	51.5	56.4	61.3	66.1	71.0	75.9
	Gmelina- mango - pigeon pea	27.3	32.7	38.1	43.5	48.9	54.3	59.7	65.1	70.5	75.9	81.3
Available Mn (mg kg^{-1})	*Gmelina*	78.9	79.7	80.5	81.3	82.1	83.0	83.8	84.6	85.4	86.2	87.0
	Mango	78.9	80.7	82.4	84.2	86.0	87.8	89.5	91.3	93.1	94.8	96.6
	Pigeon pea	78.8	83.1	87.3	91.6	95.8	100.1	104.4	108.6	112.9	117.1	121.4

	Gmelina- mango	79.0	82.5	86.0	89.5	93.0	96.5	99.9	103.4	106.9	110.4	113.9
	Gmelina- pigeon pea	78.8	82.5	86.2	89.9	93.6	97.3	100.9	104.6	108.3	112.0	115.7
	Mango - pigeon pea	79.0	83.8	88.5	93.3	98.0	102.8	107.5	112.3	117.0	121.8	126.5
	Gmelina- mango - pigeon pea	79.1	84.6	90.0	95.5	101.0	106.5	111.9	117.4	122.9	128.3	133.8
Available Cu (mg kg^{-1})	*Gmelina*	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.4	3.5
	Mango	2.4	2.6	2.8	2.9	3.1	3.3	3.5	3.6	3.8	4.0	4.2
	Pigeon pea	2.5	2.6	2.8	2.9	3.1	3.2	3.3	3.5	3.6	3.8	3.9
	Gmelina- mango	2.4	2.6	2.8	3.0	3.2	3.3	3.5	3.7	3.9	4.1	4.3
	Gmelina- pigeon pea	2.5	2.8	3.0	3.3	3.6	3.9	4.1	4.4	4.7	4.9	5.2
	Mango - pigeon pea	2.4	2.7	3.0	3.3	3.6	3.9	4.2	4.5	4.8	5.1	5.4
	Gmelina- mango - pigeon pea	2.3	2.6	2.9	3.3	3.6	3.9	4.2	4.5	4.9	5.2	5.5
Available Zn (mg kg^{-1})	*Gmelina*	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	Mango	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4
	Pigeon pea	0.2	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.6	0.6	0.7
	Gmelina- mango	0.2	0.3	0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7
	Gmelina- pigeon pea	0.2	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.6	0.6	0.7
	Mango - pigeon pea	0.2	0.3	0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7
	Gmelina- mango - pigeon pea	0.2	0.3	0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7

Supplementary Table S9

Earthworm density in the soil under different production systems after 10 years

Production system	Earthworm density (populationm^{-2})[1]					
	LM	*MP*	*PEx*	*DN*	*EO*	*Pee*
Gmelina	33	13	8	0.68	0.48	0.66
Mango	29	10	5	0.45	0.31	0.00
Gmelina-mango	43	16	13	0.69	0.50	0.66
Pigeon pea	30	5	4	0.00	0.00	0.00
Gmelina-pigeon pea	43	19	9	0.68	0.50	0.66
Mango-pigeon pea	31	10	6	0.45	0.30	0.00
Gmelina-mango-pigeon pea	53	27	13	2.00	0.54	0.70

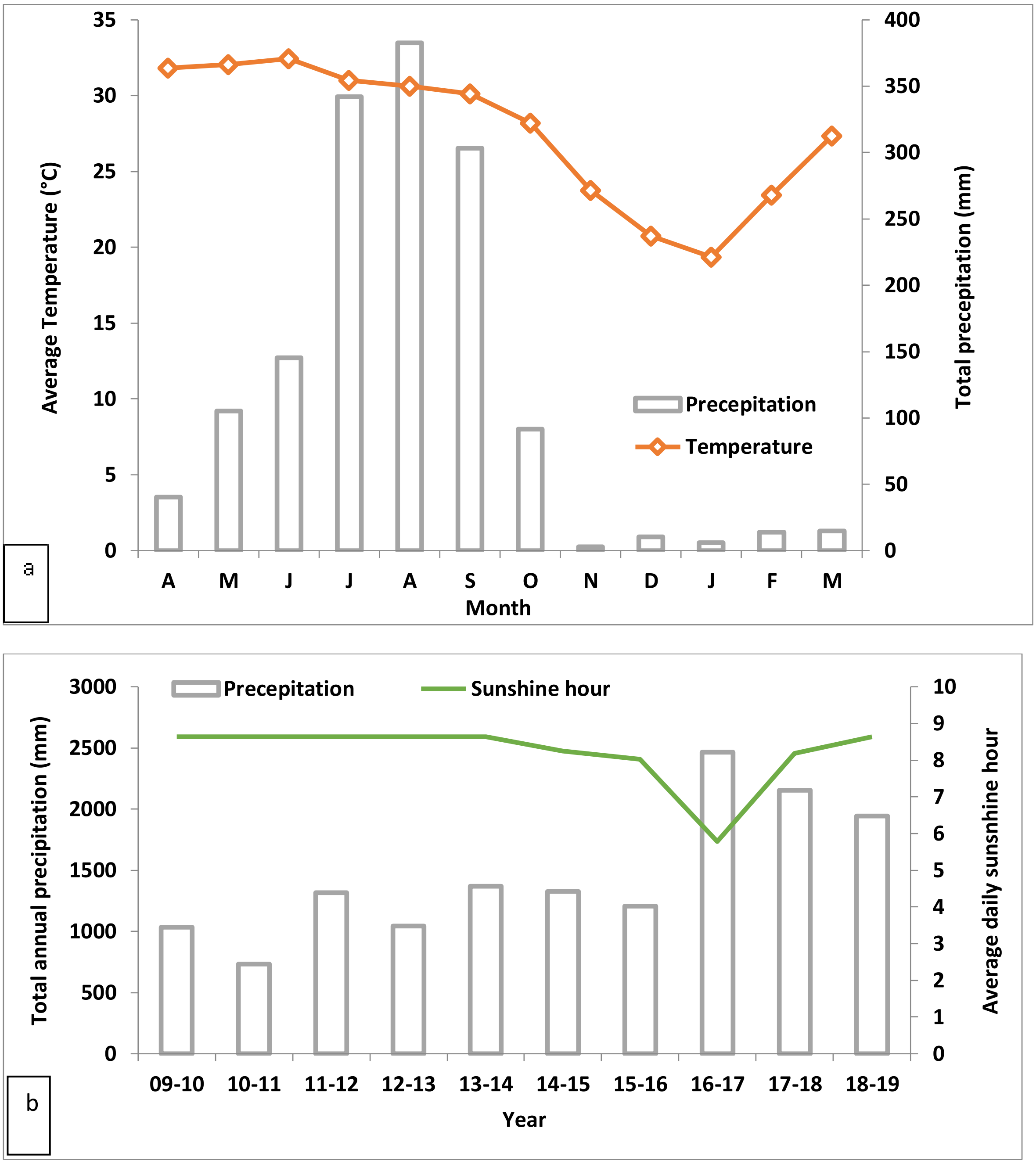
[1]*LM: Lampitomauritii, MP: Metaphireposthuma, PEx: Perionyx excavate, DN: Drawidanepalensis, EO: Eutyphoeusorientalis, PEe: Polypheretima elongate*

Supplementary Table 10

Factor loadings derived from the principal component analysis. For each variable, bold-faced values correspond to the factor for which the squared cosine is the largest.

Active variables	PC1	PC2	PC3	PC4	PC5	PC6
Microbial resilience index	**0.984**	-0.029	0.069	-0.083	0.131	-0.039
Water stable aggregates	**0.983**	-0.114	0.140	-0.026	-0.034	0.001
Net energy	**0.960**	-0.218	0.144	0.073	0.066	0.014
Dehydrogenase activity	**0.956**	-0.188	0.215	-0.031	-0.045	-0.037
Soil organic carbon	**0.948**	0.053	-0.186	0.039	-0.251	-0.006
Soil pH	**0.861**	0.206	-0.046	-0.462	0.006	0.026
Soil C sequestration	**0.855**	-0.144	-0.484	0.103	0.056	0.010
Biomass	**0.853**	-0.143	-0.488	0.105	0.054	0.008
Net present value	**0.736**	0.657	0.080	0.137	0.023	-0.004
Greenhouse gases	**0.686**	-0.545	0.466	0.116	-0.006	0.031
Simpson's diversity index	0.548	**0.798**	0.215	0.127	0.019	0.015
Eigen value	8.181	1.549	0.871	0.299	0.095	0.005
Variance explained	74.372	14.085	7.914	2.718	0.864	0.048
Cumulative variance	74.372	88.457	96.371	99.088	99.950	99.952

Supplementary Fig. S1 Location map of the experimental site (22° 43′ N, 88° 30′ E and 9.75 m above mean sea level).



Supplementary Fig S2 Monthly average temperature and monthly precipitation (a);annual precipitation and average daily sunshine hour (b) during the period of 2009-10 to 2018-19.

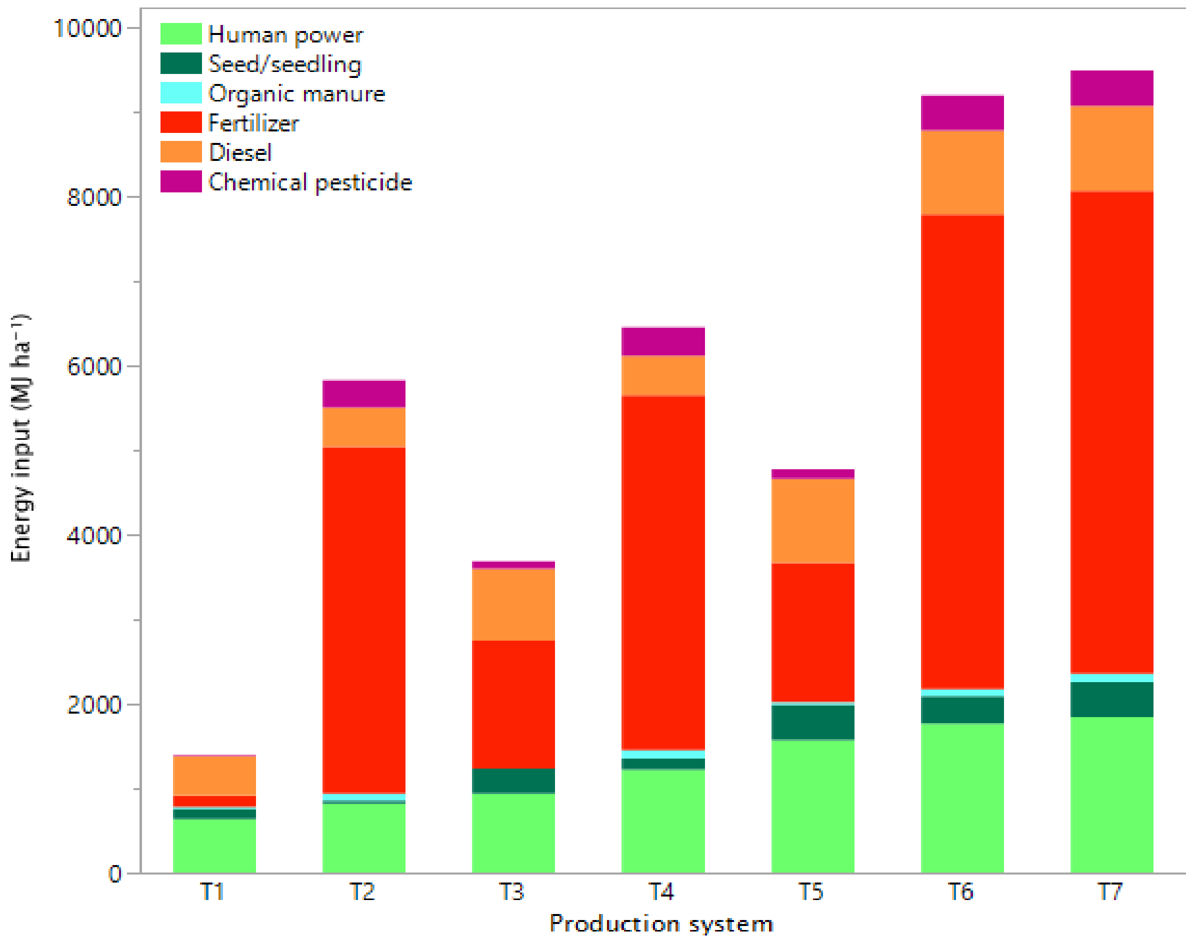


Fig. S3 Annual average energy input in different production systems under study (2009-10 to 2018-19)

T1: *Gmelina*, T2: Mango, T3: Pigeon pea, T4: *Gmelina*-mango, T5: *Gmelina*-pigeon pea, T6: Mango-pigeon pea, and T7: *Gmelina*-mango-pigeon pea